PHYS 4027 / 5027 · FALL 2026 · LECTURE 20

Conservation laws and the finite-volume method

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Last time

- The diffusion equation. FTCS is stable for $\alpha = D\Delta t/\Delta x^2 \leq \frac{1}{2}$, but that time step is tiny.
- Implicit FTCS is unconditionally stable, at the price of a tridiagonal solve per step. Crank-Nicolson is also second order in time.
- Before that, three schemes for the advection equation, upwind, Lax-Friedrichs and Lax-Wendroff, and the CFL condition $v\Delta t \leq \Delta x$.

Today: conservation laws, the finite-volume method, and the scheme of Project 2.

Part I · Conservative form

Conservative form

Burgers' equation (Lecture 17):

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0.$$

Since $u\,\partial_x u = \partial_x(u^2/2)$, the same equation is

$$\frac{\partial u}{\partial t} + \frac{\partial F}{\partial x} = 0, \qquad F = \frac{u^2}{2}.$$

This is the *conservative form*, $\partial_t Q + \nabla\cdot\mathbf{F} = 0$: Q is a density, $\mathbf{F}$ its flux. The continuity equation $\partial_t \rho + \nabla\cdot(\rho\mathbf{v}) = 0$ is one.

The integral form

Integrate over $[x_1, x_2]$:

$$\frac{d}{dt}\int_{x_1}^{x_2} Q\,dx = F(x_1) - F(x_2).$$

- Q changes only by what flows through the ends. On a periodic box the total is constant.
- No derivatives of Q: it holds across a jump, and gives the shock speed $\dot{x}_s = (F_L - F_R)/(Q_L - Q_R)$ (Lecture 17).
- u and u^2 both satisfy a conservation law where u is smooth, with different shock speeds. The physics decides which is conserved.

On a grid

Cells of width Δx, the cell *average* Q_j, faces at $x_{j\pm1/2}$. The integral form over one cell and one step:

$$Q_j^{n+1} = Q_j^n - \frac{\Delta t}{\Delta x}\left(F_{j+1/2} - F_{j-1/2}\right).$$

- $F_{j+1/2}$ is the flux through the face between cells j and $j+1$: what leaves j enters $j+1$.
- Sum over j and the fluxes cancel in pairs. $\sum_j Q_j \Delta x$ is conserved to rounding, for any $F_{j+1/2}$.
- Such a scheme also gets the shock speed right, since that followed from conservation alone.

Part II · The finite-volume method

Finite volume method

The finite volume method exploits this structure, allowing us to write the update as:

$$\begin{aligned}\int_{\text{cell}} Q^{n+1}\,\mathrm{d}V &= \int_{\text{cell}} Q^{n}\,\mathrm{d}V - \Delta t \int_{\text{cell}} \nabla\cdot\mathbf{F}\,\mathrm{d}V \\ &= \int_{\text{cell}} Q^{n}\,\mathrm{d}V - \Delta t \int_{\text{boundary}} \mathbf{F}\cdot\mathrm{d}\mathbf{S}\end{aligned}$$

Define $\bar{Q}$ as the average Q in each cell, we can write:

$$\bar{Q}^{n+1}\,\Delta V = \bar{Q}^{n}\,\Delta V - \Delta t \sum_{\text{faces}} \mathbf{F}\cdot\mathbf{S}$$

ρ by itself is a conserved quantity, and $\int_{\text{cell}} \rho\,\mathrm{d}V$ is the mass in the cell. This is what we use as one of the conserved quantities in Project 2.

Face fluxes

The update is exact. A scheme is a formula for $F_{j+1/2}$ from the two cells sharing the face, $Q_L = Q_j$ and $Q_R = Q_{j+1}$.

Upwind, flow to the right: $F_{j+1/2} = F(Q_L)$.

Lax-Friedrichs:

$$F_{j+1/2} = \frac{1}{2}\left[F(Q_L) + F(Q_R)\right] - \frac{\Delta x}{2\Delta t}\left(Q_R - Q_L\right).$$

In the update this gives the scheme of Lecture 18; the second term is its numerical diffusion, $D = \Delta x^2/2\Delta t$.

The Rusanov flux

There are many ways to construct the fluxes at the cell faces. One way is the Rusanov flux:

$$F = \frac{1}{2}(F_L + F_R) - \frac{v_{\max}}{2}(Q_R - Q_L).$$

Subscript L and R are quantities calculated using the fluid variables in the cell to the left and right of the face, and $v_{\max}$ is the largest speed at which information travels across the face. This is what we will be using in Project 2.

Lax-Friedrichs with $\Delta x/\Delta t$ replaced by $v_{\max} \leq \Delta x/\Delta t$: less diffusion, $v_{\max}\Delta x/2$ instead of $\Delta x^2/2\Delta t$.

Part III · The scheme of Project 2

Project 2

In Project 2, we will solve the equations for an ideal fluid:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$
$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{1}{\rho}\nabla P$$

These are called "Euler equations". They are the limit of Navier-Stokes equations when fluid viscosity becomes zero.

The velocity equation is not in conservation form. Momentum $\rho\mathbf{v}$ and energy $U = \rho(\frac{1}{2}v^2 + u)$ are conserved; their equations are.

A system of conservation laws

In 1D, the Euler equations in conservative form are $\partial_t Q + \partial_x F = 0$ with

$$Q = \begin{pmatrix} \rho \\ \rho v \\ U \end{pmatrix}, \qquad F = \begin{pmatrix} \rho v \\ \rho v^2 + P \\ (U + P)\, v \end{pmatrix}, \qquad U = \frac{1}{2}\rho v^2 + \frac{P}{\gamma - 1}.$$

Part II applies to the vector Q component by component. Mass, momentum and energy are conserved to rounding.

In 2D there are two flux vectors, F_x and F_y; the handout lists them.

Signal speeds and the time step

A gas carries information at three speeds: the flow speed v, and sound at $v \pm c_s$,

$$c_s = \sqrt{\gamma P / \rho}.$$

The CFL condition with the fastest signal on the grid, set every step:

$$\Delta t = C_{\mathrm{CFL}} \, \frac{\min(\Delta x, \Delta y)}{\max_{ij} \left(c_s + |\mathbf{v}| \right)}.$$

On a face, $c_s + |\mathbf{v}|$ is the $v_{\max}$ of the Rusanov flux.

$C_{\mathrm{CFL}} \le 1$ is not enough in 2D; the handout uses 0.4.

Second order

$Q_L = Q_j, Q_R = Q_{j+1}$ is piecewise constant: first order in space, diffusion $\propto \Delta x$. The handout's scheme:

- Half step: advance ρ, $\mathbf{v}$, P by $\Delta t/2$ with centered differences (the midpoint rule, Lecture 6). Second order in time.
- Reconstruct: extrapolate to the faces with the centered slope, $\rho_j^R = \rho_j + \frac{1}{4}(\rho_{j+1} - \rho_{j-1})$. Piecewise linear. Second order in space.
- Rusanov flux from the two face values; update Q; recover ρ, $\mathbf{v}$, P.

Two dimensions

Cells $\Delta x \times \Delta y$ indexed by (i, j), four faces each:

$$Q_{ij}^{n+1}\,\Delta x \Delta y = Q_{ij}^{n}\,\Delta x \Delta y - \left(F_{x,ij} - F_{x,i-1,j}\right)\Delta y\,\Delta t - \left(F_{y,ij} - F_{y,i,j-1}\right)\Delta x\,\Delta t.$$

$F_{x,ij}$ is the x-flux through the right face of cell (i, j), from the two cells on either side as in 1D. The y-faces use F_y.

Periodic boundaries: a layer of ghost cells copying the far side. Data layout, loops and output are the agent's part of the project.

The CFL condition in 2D

Why is $C_{\mathrm{CFL}} = 1$ not safe in 2D? The simplest case: 2D advection, $\partial_t u + \mathbf{v} \cdot \nabla u = 0$, with Lax-Friedrichs, now averaging four neighbours:

$$u_{j,k}^{n+1} = \frac{1}{4}\left(u_{j+1,k}^{n} + u_{j-1,k}^{n} + u_{j,k+1}^{n} + u_{j,k-1}^{n}\right) - \frac{v_x \Delta t}{2\Delta x}\left(u_{j+1,k}^{n} - u_{j-1,k}^{n}\right) - \frac{v_y \Delta t}{2\Delta y}\left(u_{j,k+1}^{n} - u_{j,k-1}^{n}\right).$$

Von Neumann analysis with a 2D mode $u_{j,k}^{n} = \xi^n e^{i(k_x j \Delta x + k_y k \Delta y)}$, for $\Delta x = \Delta y = \Delta$:

$$\Delta t \le \frac{\Delta}{\sqrt{2}\,|\mathbf{v}|}.$$

An extra $\sqrt{2}$ compared with 1D; $\sqrt{d}$ in d dimensions.

Lax-Friedrichs scheme in 2D

Demo of the Lax-Friedrichs scheme in 2D: a Gaussian on a 128×128 periodic grid, $\mathbf{v} = (1.5, 1)$, with Δt at the 2D CFL limit.



The peak drops from 1 to 0.36 during the run: in 2D the scheme is diffusive even at the CFL limit. The smearing is across the direction of motion; along it the profile keeps its width.

Part IV · Checking a fluid code

What your agent cannot do for you

Three checks

- Conservation. Sum ρ, ρv_x, ρv_y and U over all cells at every snapshot. On a periodic box each total is constant to rounding, a relative change of order 10^{-14}.
- Symmetry. A symmetric initial condition stays symmetric.
- Resolution. Run 128^2, 256^2 and 512^2. What the equations determine settles as the grid is refined.

A sound wave

The handout's test. Perturb a uniform gas, $\rho = \rho_0 + \rho_1$, $P = P_0 + P_1$, $\rho_1 \ll \rho_0$. To first order:

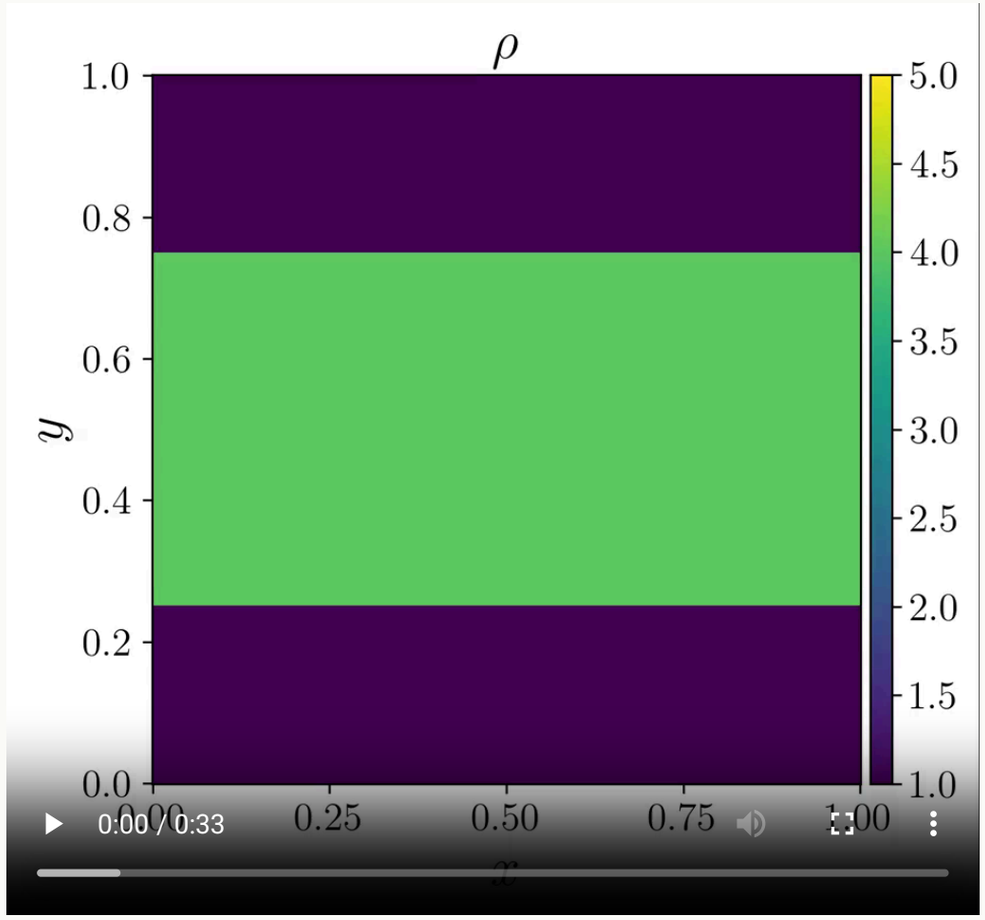
$$\frac{\partial^2 P_1}{\partial t^2} - c_s^2 \, \frac{\partial^2 P_1}{\partial x^2} = 0, \qquad c_s = \sqrt{\frac{\gamma P_0}{\rho_0}}.$$

A sine wave with $P_1 = c_s^2 \rho_1$ and $v_x = c_s \rho_1 / \rho_0$ travels to the right at c_s with constant amplitude.

Measure the speed, and the decay of the amplitude at two resolutions.

Kelvin-Helmholtz instability

The Kelvin-Helmholtz instability occurs at an interface in a fluid where two sides move at different velocities:



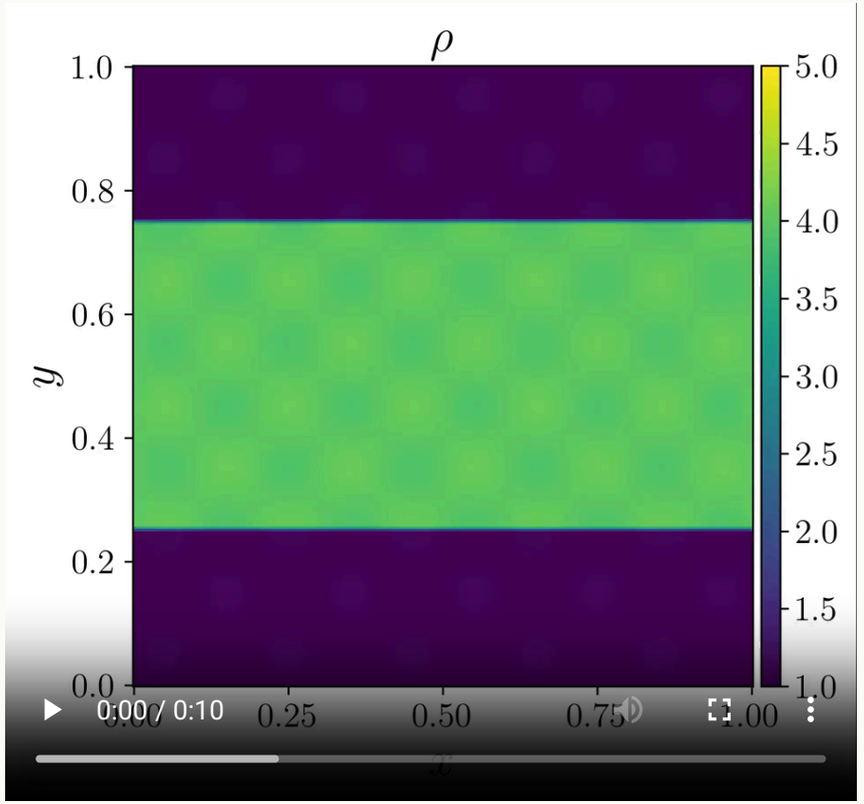
Kelvin-Helmholtz instability

The KH instability is ubiquitous in nature:



Why finite volume?

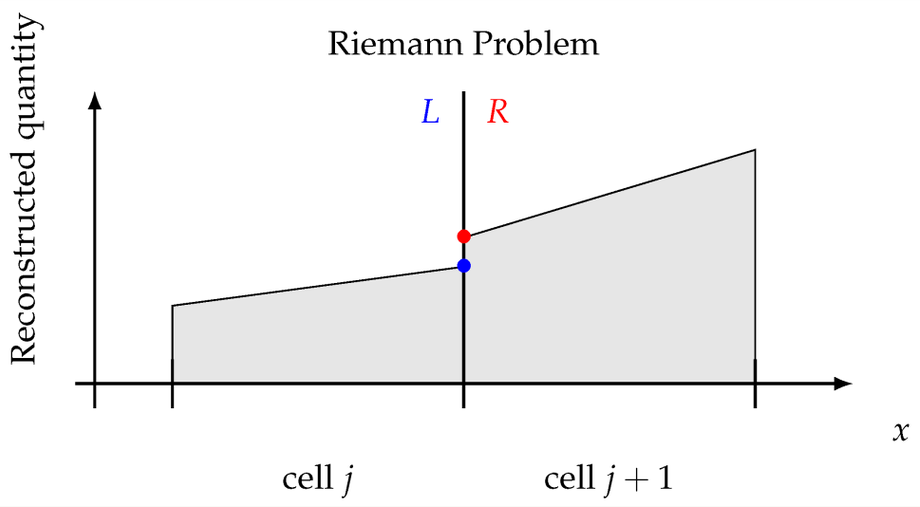
This is the result from the usual Lax-Friedrichs scheme on the same problem:



Numerical diffusion is so strong that it completely overwhelms the instability.

Riemann solvers

To capture discontinuities in the fluid, an even better way is to introduce a *Riemann solver* at each cell boundary.



Rusanov is the crudest approximate Riemann solver: one speed for all three waves. The 5027 part of the handout adds HLLC.

Summary

- Conservative form: the integral form holds across jumps and fixes the shock speed.
- Finite volume: cell averages, face fluxes, conservation to rounding. The scheme is the flux formula: Lax-Friedrichs, or Rusanov with the local signal speed.
- Project 2: Rusanov with $c_s + |\mathbf{v}|$, a half step and a linear reconstruction, the 2D CFL condition.
- Checks: conservation, symmetry, resolution, the sound wave.

Next time: Poisson's equation and relaxation methods.